1.38MW Kohat tunnel, kohat tunnel

Report

Project Name	Kohat tunnel
Project Address	kohat tunnel
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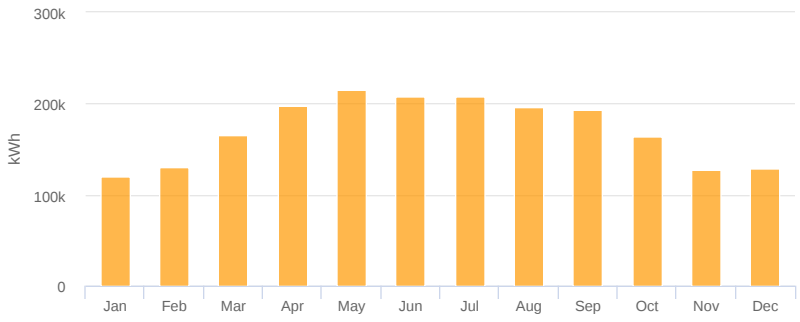
System Metrics

Design	1.38MW
Module DC Nameplate	1.38 MW
Inverter AC Nameplate	1.50 MW Load Ratio: 0.92
Annual Production	2.052 GWh
Performance Ratio	82.8%
kWh/kWp	1,483.9
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)
Simulator Version	bfabe04d4f-96fc2a8ce2-9340f8b948-7b59da0912

Project Location

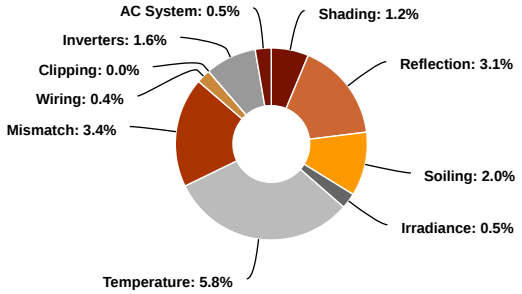


Monthly Production



Month	GHI (kWh/m²)	POA (kWh/m²)	Shaded (kWh/m²)	Nameplate (kWh)	Grid (kWh)
January	78.8	100.0	98.9	129,136.5	119,948.0
February	91.2	108.3	107.2	140,745.6	129,596.2
March	127.1	141.2	139.6	183,221.3	165,007.7
April	163.3	172.0	170.0	223,589.6	197,039.0
May	190.5	192.5	190.4	250,453.8	215,832.4
June	188.3	187.2	184.8	242,890.9	208,230.5
July	185.8	186.0	183.7	241,553.0	207,679.8
August	168.7	174.9	172.7	227,227.8	196,238.5
September	155.9	170.9	169.1	222,544.7	193,226.6
October	122.4	142.2	140.7	184,873.3	163,192.5
November	86.8	108.7	107.4	140,378.5	127,575.5
December	81.7	108.3	107.0	139,734.4	128,637.2

Sources of System Loss



Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,640.7	
	POA Irradiance	1,792.2	9.2%
	Shaded Irradiance	1,771.3	-1.2%
	Irradiance after Reflection	1,716.3	-3.1%
	Irradiance after Soiling	1,682.0	-2.0%
	Total Collector Irradiance	**1,681.9**	**0.0%**
Energy (kWh)	Nameplate	2,326,349.4	
	Output at Irradiance Levels	2,315,009.2	-0.5%
	Output at Cell Temperature Derate	2,180,253.8	-5.8%
	Output After Mismatch	2,105,362.1	-3.4%
	Optimal DC Output	2,096,058.6	-0.4%
	Constrained DC Output	2,096,053.4	0.0%
	Inverter Output	2,062,516.5	-1.6%
	Energy to Grid	**2,052,203.9**	**-0.5%**
Temperature Metrics			
	Avg. Operating Ambient Temp		24.1 °C
	Avg. Operating Cell Temp		33.5 °C
Simulation Metrics			
	Operating Hours		4626
	Solved Hours		4626

Condition Set

Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type	a	b	Temperature Delta								
	Fixed Tilt	-3.56	-0.075	3°C								
	Flush Mount	-2.81	-0.0455	0°C								
	East-West	-3.56	-0.075	3°C								
	Carport	-3.56	-0.075	3°C								
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											
Module & Component Characterizations	Type	Component	Characterization									
	Module	JKM585M-78HL4-V (1500V) (Jinko)	Spec Sheet Characterization, PAN									
	Inverter	SUN2000-150K-MG0 (Huawei)	Spec Sheet									

Components

Component	Name	Count
Inverters	SUN2000-150K-MG0 (Huawei)	10 (1.50 MW)
Strings	10 AWG (Copper)	136 (27,522.0 ft)
Module	Jinko, JKM585M-78HL4-V (1500V) (585W)	2,364 (1.38 MW)

Wiring Zones

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	5-19	Along Racking

Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Fixed Tilt	Landscape (Horizontal)	Module: 15°	Module: 180°	9.0 ft	3x1	788	2,364	1.38 MW

Detailed Layout2

